EyeExplore

Visual Eyetracking Dashboard

User Guide Manual

Version 2.0

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A comprehensive tool for eye-tracking data analysis and visualization
with support for built-in datasets, custom data uploads, and K-Coefficient analysis.

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1. Introduction

EyeExplore is a web-based interactive dashboard for visualizing and analyzing eye-tracking data. It provides researchers with powerful tools to explore gaze patterns, fixation behaviors, and scanpath dynamics through an intuitive dual-panel interface. The dashboard supports multiple data sources, rich visualization types, and advanced analysis features including Areas of Interest (AOI) analysis, K-Coefficient classification, and object detection.

Key capabilities include:

- Side-by-side visualization comparison with independent panel controls
- 14 different visualization types ranging from basic stimulus display to advanced 3D scanpaths
- Support for built-in research datasets, custom data uploads, and K-Coefficient video datasets
- Interactive AOI drawing and analysis tools
- Real-time scanpath animation playback
- Object detection overlay using YOLO models
- Export and download capabilities for all visualizations

2. Getting Started

2.1 Prerequisites

EyeExplore runs as a local web application. Ensure you have Python 3.10+ installed along with the required dependencies (Dash, Plotly, NumPy, Pillow, etc.).

2.2 Launching the Dashboard

1. Open a terminal and navigate to the EyeExplore directory.
2. Run: python app.py
3. Open your web browser and navigate to: http://localhost:8050
4. The dashboard will load with the Built-in Datasets mode active by default.

2.3 System Requirements

- Modern web browser (Chrome, Firefox, Safari, or Edge recommended)
- Screen resolution of at least 1280x720 for optimal viewing
- Minimum 4 GB RAM for large dataset visualizations

3. Dashboard Overview - Built-in Datasets Mode

The Built-in Datasets mode is the default mode when you launch EyeExplore. It provides access to pre-loaded eye-tracking research datasets. The screenshot below shows the main dashboard interface with numbered callouts for each control.

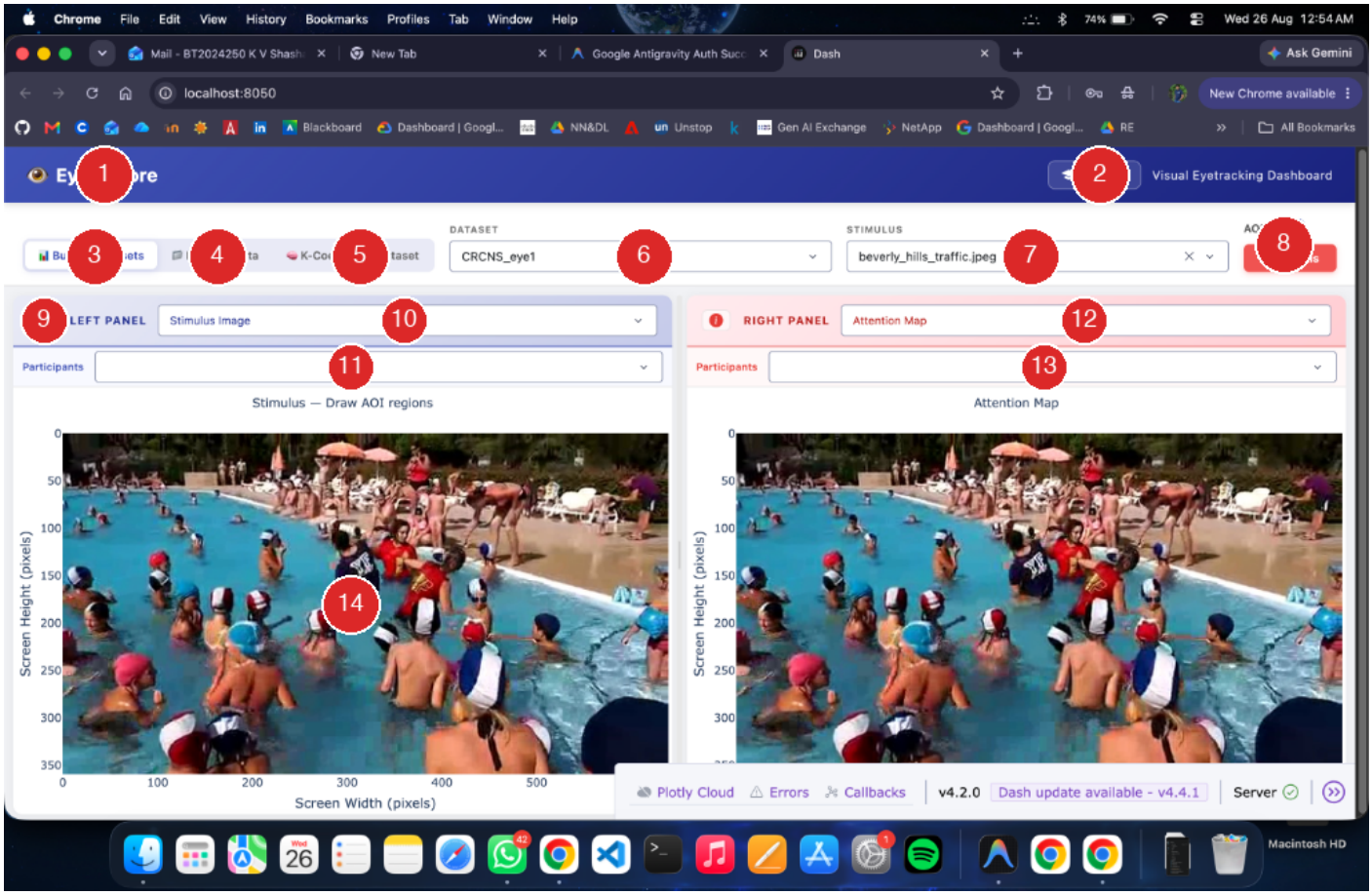


Figure 1: Built-in Datasets Mode - Main Dashboard

Control Reference

1 EyeExplore Logo - The application logo and branding. Indicates the dashboard is active and running.

2 Tutorial Button - Launches an interactive guided tour that walks you through each dashboard control step-by-step with spotlight animations and explanatory tooltips.

3 Built-in Datasets Tab - Switches to the built-in datasets mode. Pre-loaded research datasets (e.g., CRCNS_eye1) are available with stimuli and participant gaze data ready to explore.

4 My Own Data Tab - Switches to custom data upload mode. Allows you to upload your own stimulus images/videos and scanpath data files for analysis.

5 K-Coefficient Dataset Tab - Switches to the K-Coefficient analysis mode for ambient/focal attention classification using video stimuli and key frame extraction.

6 Dataset Dropdown - Select from available built-in datasets (e.g., CRCNS_eye1). Changing the dataset reloads all stimulus images and participant lists for both panels.

7 **Stimulus Dropdown** - Select which stimulus image to analyze. Both panels update to show visualizations for the selected stimulus. Each stimulus has its own set of participant recordings.

Control Reference (continued)

8 AOIs Badge & Clear AOIs - The blue badge shows the count of Areas of Interest (AOI) regions you have drawn. The red 'Clear AOIs' button removes all drawn AOI regions. AOIs are used by Scarf Plots, AOI Timelines, and Transition Matrices.

9 Info Button (i) - Opens a detailed modal explaining the currently selected visualization type - what it represents, how to interpret it, and usage tips. Available for both left (blue) and right (red) panels.

10 Left Panel - Visualization Type - Dropdown to select the visualization for the left panel. 14 different visualization types are available (see Section 6). The left panel header is styled in blue.

11 Left Panel - Participants - Multi-select dropdown to choose which participants' eye-tracking data to display in the left panel. You can select multiple participants for aggregate views (heatmaps, scarf plots) or individual ones for scanpath overlays.

12 Right Panel - Visualization Type - Independent dropdown for the right panel (styled in red). Select a different visualization to compare side-by-side with the left panel.

13 Right Panel - Participants - Participant selector for the right panel. You can compare different participant groups across panels to identify behavioral differences.

14 Visualization Canvas - The main plotting area where visualizations are rendered. Interactive features include: zoom (scroll wheel), pan (click and drag), draw AOI regions (use the drawing tools in the toolbar), hover for detailed tooltips, and download as PNG.

4. Dashboard Overview - My Own Data Mode

The 'My Own Data' mode allows you to upload your own stimulus images/videos and scanpath files for analysis. This is ideal for researchers who want to analyze their own experimental data using EyeExplore's visualization tools.

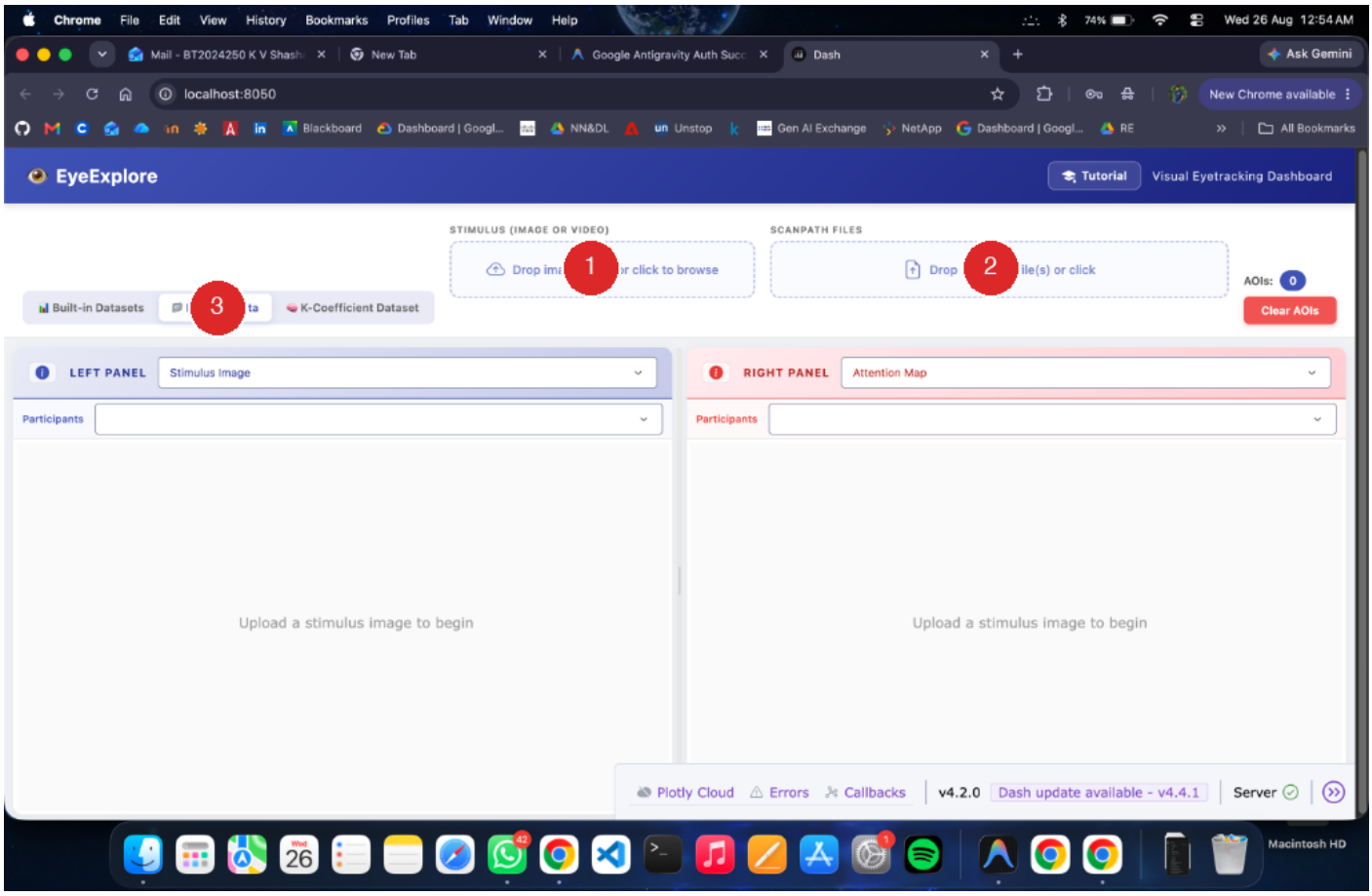


Figure 2: My Own Data Mode - Upload Interface

Control Reference

1 **Stimulus Upload Area** - Drag and drop or click to browse for your stimulus image or video file. Supported formats include JPEG, PNG, and common video formats. The uploaded stimulus will be displayed in both panels once loaded.

2 **Scanpath Files Upload** - Upload one or more scanpath data files. These files contain gaze coordinates (x, y positions) and fixation data for each participant. Each file represents one participant's eye-tracking recording for the uploaded stimulus.

3 **My Own Data Tab (Active)** - This tab is highlighted when active, indicating custom data mode. The dataset and stimulus dropdowns from Built-in mode are replaced with upload areas.

After uploading both a stimulus and scanpath files, all visualization types become available just as they are with built-in datasets. You can use the same dual-panel layout, participant selectors, and AOI drawing tools.

5. Dashboard Overview - K-Coefficient Dataset Mode

The K-Coefficient Dataset mode provides access to a specialized dataset for ambient/focal attention analysis. This mode features video stimuli with key frame extraction and K-Coefficient-based gaze classification.

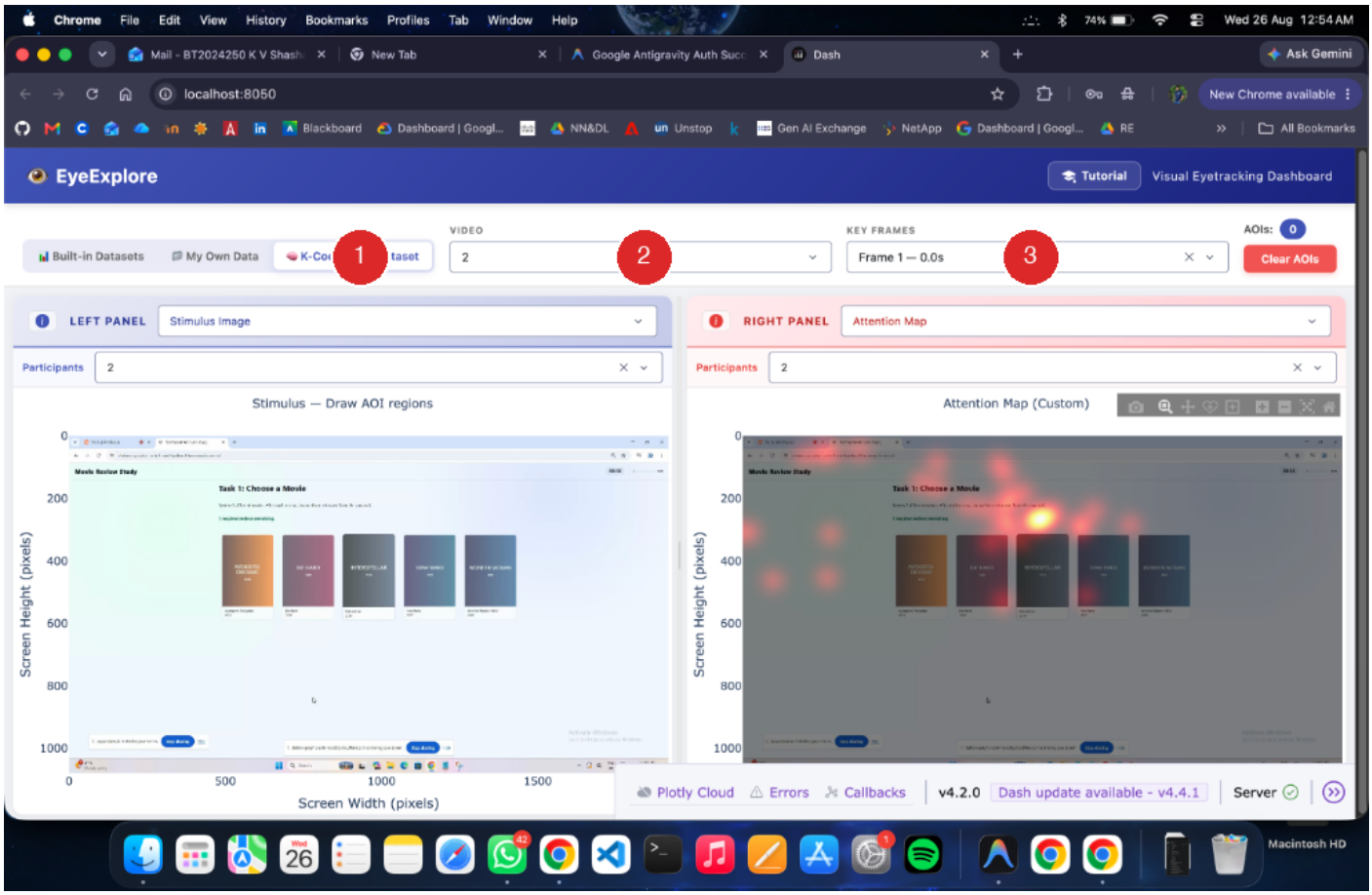


Figure 3: K-Coefficient Dataset Mode

Control Reference

1 **K-Coefficient Dataset Tab (Active)** - Highlighted when active. This mode loads a specialized dataset with 26 participants who viewed video stimuli. The interface adapts to show video-specific controls.

2 **Video Dropdown** - Select which video stimulus to analyze. Videos are identified by participant numbers. The selected video's key frames and gaze data are loaded automatically.

3 **Key Frames Dropdown** - Select specific key frames extracted from the video for analysis. Each key frame captures a moment in the video where significant gaze events occurred. Frame timestamps are shown for reference (e.g., 'Frame 1 - 0.0s').

In K-Coefficient mode, additional visualization types become especially useful:
- K-Colored Scanpath: Overlays gaze points colored by their K-Coefficient value
- K-Coefficient Timeline: Shows the K value over time for ambient/focal classification
- Focal/Ambient Heatmap: Separate heat maps for focal and ambient fixations

6. Visualization Types

EyeExplore provides 14 different visualization types, accessible from the panel dropdowns. Each visualization offers a unique perspective on eye-tracking data.

- **Stimulus Image:** Displays the raw stimulus image with AOI drawing support. Use the plot toolbar to draw rectangular or free-form AOI regions directly on the image. This is the foundation view for defining analysis regions.
- **Scanpath Animation:** Animated playback of participant gaze paths over the stimulus. Watch fixation points appear in sequence with saccade connections. Useful for understanding temporal viewing patterns and search strategies.
- **Scanpath Overlay:** Static overlay of complete scanpaths on the stimulus. Shows fixation points (circles sized by duration) connected by saccade lines. Select multiple participants to compare viewing patterns simultaneously.
- **Attention Map:** Heat map showing aggregate fixation density across the stimulus. Hot regions (red/yellow) indicate areas that attracted the most visual attention. Cool regions (blue/green) received less attention. Excellent for identifying salient regions.
- **Scarf Plot:** Horizontal bar chart showing the temporal sequence of AOI visits for each participant. Each colored segment represents time spent in a specific AOI. Requires AOI regions to be drawn first. Great for comparing viewing sequences.
- **AOI Timeline:** Time-series line chart showing when each AOI was visited and for how long. Tracks transitions between AOI regions over the entire viewing duration. Requires AOI regions to be defined.
- **3D Scanpath:** Three-dimensional visualization where X and Y represent gaze coordinates and Z represents time. Shows the scanpath as a trajectory through space-time. Rotate and zoom to explore viewing patterns from different angles.

Visualization Types (continued)

- **3D Scatter:** Scatter plot in 3D space with fixation points plotted at (X, Y, Time). Point size encodes fixation duration. Helps identify fixation clusters and temporal patterns in gaze behavior.
- **Co-association Matrix:** Matrix showing the similarity of gaze patterns between pairs of participants. Darker cells indicate higher gaze agreement. Useful for identifying participant groups with similar viewing strategies.
- **Object Detection:** Applies YOLO object detection to the stimulus image and overlays detected objects with bounding boxes and confidence scores. Helps relate gaze patterns to semantic content in the image.
- **Transition Matrix:** Square matrix showing the frequency of gaze transitions between AOI regions. Cell values indicate how often gaze moved from one AOI to another. Requires AOI regions to be defined. Useful for understanding scanning strategies.
- **K-Colored Scanpath:** Scanpath overlay where each fixation point is colored according to its K-Coefficient value. Blue indicates ambient (dispersed) attention, red indicates focal (concentrated) attention. Best used with K-Coefficient datasets.
- **K-Coefficient Timeline:** Line chart showing the K-Coefficient value for each fixation over time. Values above 0 indicate focal attention; values below 0 indicate ambient attention. Tracks attention mode transitions during viewing.
- **Focal/Ambient Heatmap:** Dual heat map separating fixations into focal and ambient categories based on K-Coefficient thresholds. Shows where concentrated vs. dispersed attention occurred on the stimulus.

7. Areas of Interest (AOIs)

Areas of Interest (AOIs) are user-defined regions on the stimulus image that represent meaningful areas for analysis. Several visualization types (Scarf Plot, AOI Timeline, Transition Matrix) require AOIs to be defined.

7.1 Drawing AOIs

1. Select 'Stimulus Image' as the visualization type in either panel.
2. Use the drawing tools in the Plotly toolbar (top-right corner of the plot):
   - Rectangle tool: Draw rectangular AOI regions
   - Free-form tool: Draw custom-shaped AOI regions
3. The AOI count badge updates automatically as you draw regions.
4. Each AOI is automatically assigned a unique color for identification.

7.2 Managing AOIs

- The blue AOI badge in the top toolbar shows the current count of AOIs.
- Click 'Clear AOIs' (red button) to remove all drawn AOI regions.
- AOIs are shared between both panels - drawing in one panel applies to both.
- AOIs persist until cleared, even when switching visualization types.

8. Dual-Panel Comparison

EyeExplore's signature feature is its dual-panel layout, allowing side-by-side comparison of different visualizations or participant groups.

8.1 Independent Controls

Each panel has its own:
- Visualization type dropdown (choose different plot types per panel)
- Participant selector (compare different participant groups)
- Info button (get help specific to each panel's visualization)

For example, display a Stimulus Image in the left panel and an Attention Map in the right panel to see which areas attracted the most fixations.

8.2 Resizable Panels

A drag divider separates the two panels. Click and drag it to resize panels - make one larger to focus on a specific visualization, or keep them equal for balanced comparison.

9. Plot Toolbar & Interactions

Each visualization canvas includes a Plotly toolbar with interactive controls:

- **Zoom:** Use the scroll wheel to zoom in/out, or use the zoom-in/zoom-out buttons in the toolbar.
- **Pan:** Click and drag to pan across the visualization when zoomed in.
- **Box Select:** Draw a box to select and zoom into a specific region.
- **Lasso Select:** Free-form selection tool for identifying specific data points.
- **Draw Shapes:** Rectangle and free-form drawing tools for creating AOI regions.
- **Reset Axes:** Double-click on the plot or use the home button to reset to the default view.
- **Download:** Click the camera icon to download the current visualization as a PNG image.
- **Hover Tooltips:** Hover over data points to see detailed information (fixation duration, coordinates, participant ID, etc.).

10. Tips & Best Practices

Effective Analysis Workflow

1. Start by selecting a dataset and stimulus image.
2. View the raw Stimulus Image first to understand the visual content.
3. Draw AOI regions on areas you want to analyze (if needed).
4. Compare visualizations side-by-side:
   - Stimulus Image + Attention Map to see where people looked
   - Scanpath Overlay + Scarf Plot to analyze temporal viewing patterns
   - K-Colored Scanpath + K-Coefficient Timeline for attention mode analysis
5. Use the Info button for each visualization to understand interpretation.
6. Download visualizations as PNG for reports and presentations.

Performance Tips

- Select fewer participants for smoother animation in Scanpath Animation mode.
- Use Chrome or Firefox for the best rendering performance.
- Large datasets may take a moment to process - allow time for complex visualizations.
- Clear unused AOIs to reduce computational overhead for AOI-based plots.

Troubleshooting

- If a visualization appears blank, ensure participants are selected from the dropdown.
- AOI-based plots (Scarf, Timeline, Transition) require at least one AOI to be drawn.
- If the server is unresponsive, refresh the page or restart with 'python app.py'.
- Port conflicts can be resolved by stopping other processes on port 8050.